2026 ASA SBSS STUDENT PAPER AWARD | IMS HANNAN GRADUATE STUDENT TRAVEL AWARD

Probabilistic Symbolic Regression for Equation Discovery via Operator-induced and Regularized Symbolic Forests

Submitted to the *Journal of Machine Learning Research*

Somjit Roy

Statistics Ph.D. student

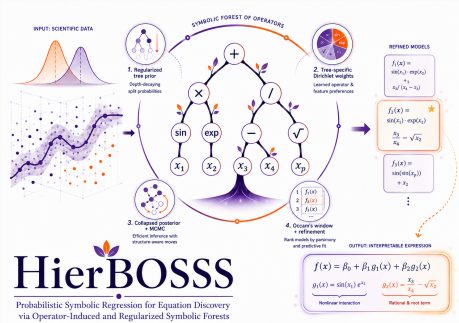
Joint work with

Pritam Dey, Bani K. Mallick, & Debdeep Pati

Texas A&M Statistics | UW-Madison Statistics

Invited Talk

SAMBA 2026 / Santos, São Paulo, Brazil



TEXAS A&M UNIVERSITY

ROADMAP

Agenda

1 Discovering Laws From Data

2 Symbolic Regression

3 Introducing `HierBOSSS`

4 What The Posterior Learns

5 Rediscovering Physics

6 Discovery of Materials Genes

7 Outlook

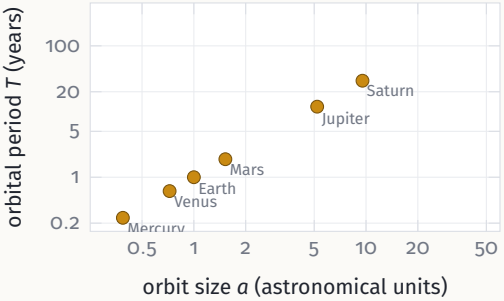
TOC

1

Discovering Laws From Data

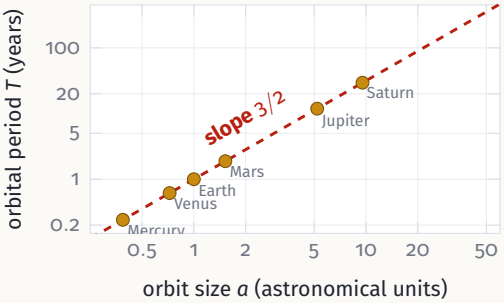
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Tycho Brahe's planetary observations $\longrightarrow$ Kepler's third law (1619)



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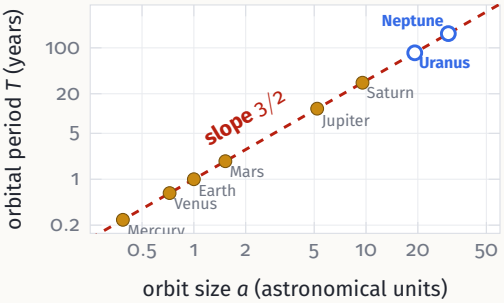
$$T^2 \propto a^3$$

one exponent explains every planet

- **Compact:** a law, not a lookup table.

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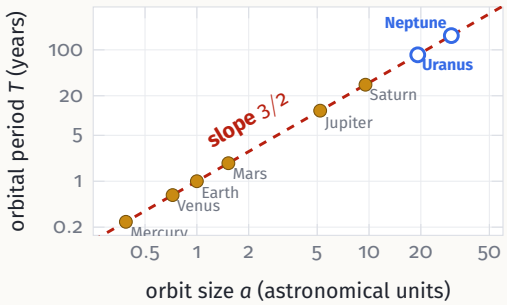
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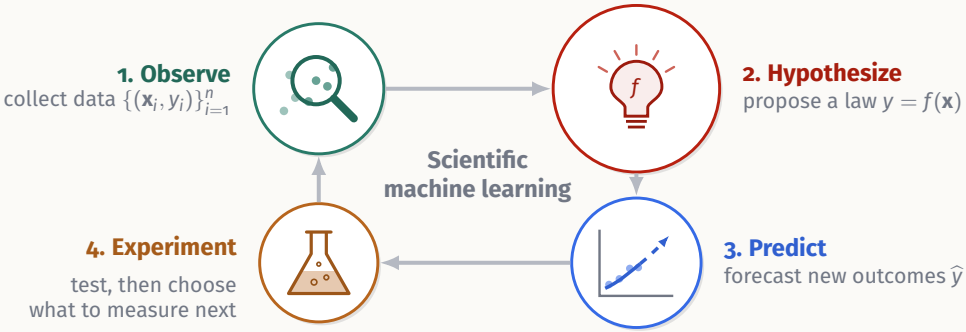
one exponent explains every planet

- **Compact:** a law, not a lookup table.
- **Extrapolates:** Uranus (1781) and Neptune (1846), found later, lie on the same line.
- **Hard to find:** Kepler took years, by hand.

Symbolic regression asks a machine to do Kepler's search: *find the equation behind the data.*

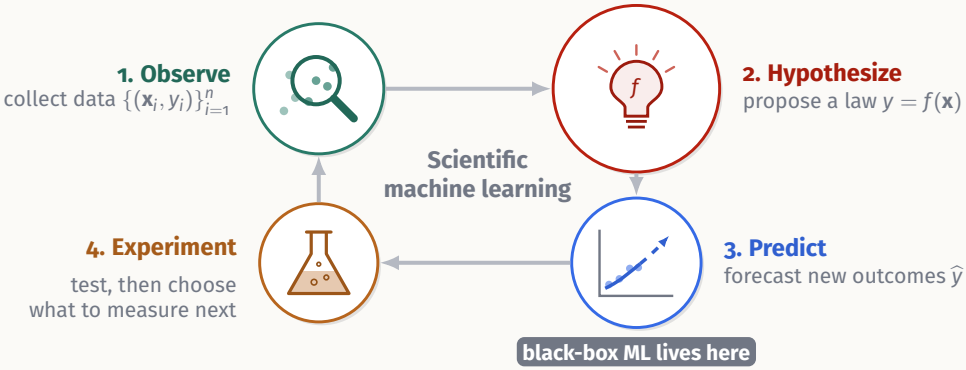
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Scientific machine learning aims to close this loop autonomously



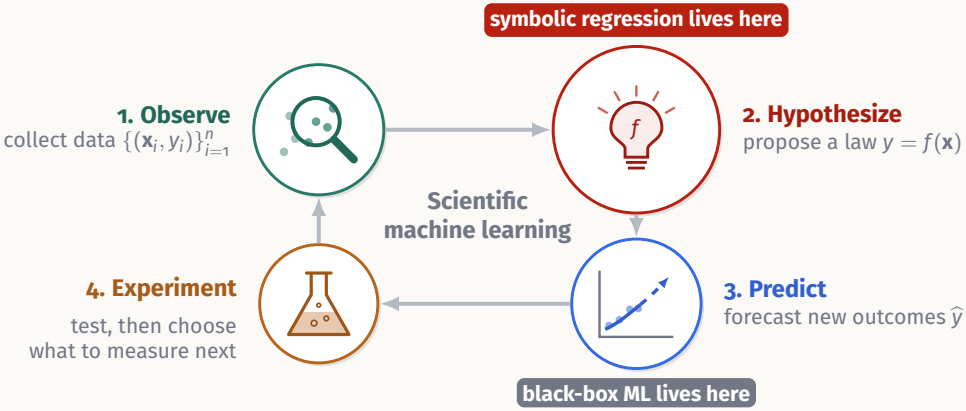
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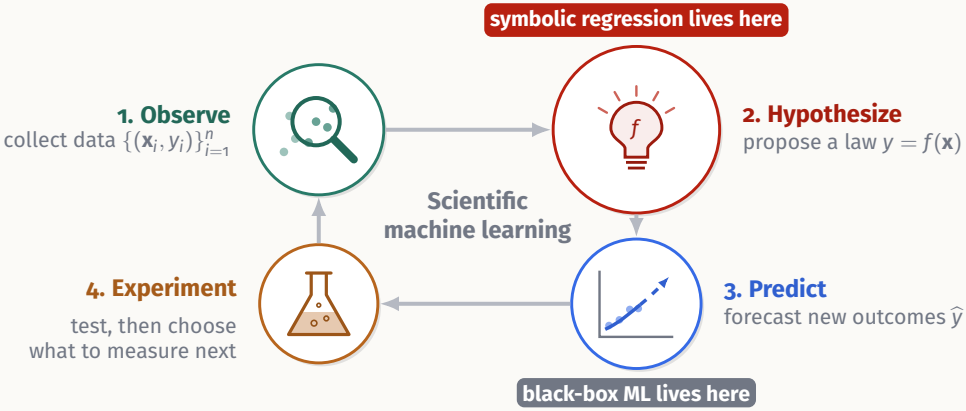
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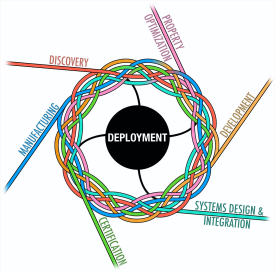
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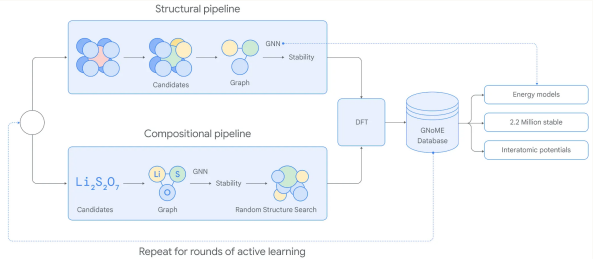


Prediction tells us *what* will happen. An equation tells us *why*, and *what to test next*.

1 AI is accelerating materials discovery



Materials Genome Initiative
Computation, experiments and data in one loop
[NSTC (2011)]



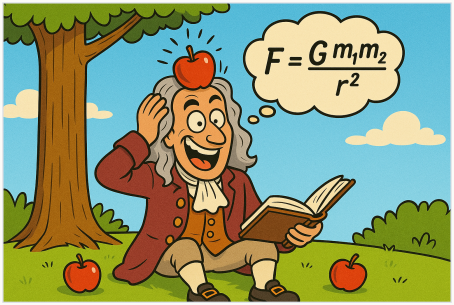
GNoME: graph networks for materials exploration
Graph networks propose crystals, physics checks them, active learning closes the loop: about 2.2 million new crystal structures predicted.
[Merchant et al., *Nature* (2023)]

Fast at *finding* candidates. But *which* physical factors make a material good?

1 AI is accelerating materials discovery

Newton had an apple. We have data.

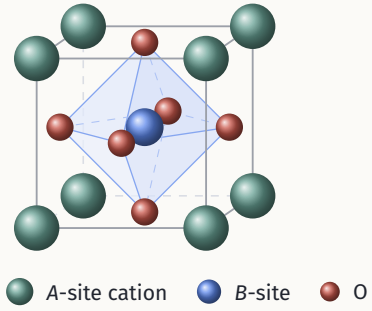
Can AI discover new materials
and recover the interpretable laws that govern them?



Fast at *finding* candidates. But *which* physical factors make a material good?

1 The motivating problem

Descriptor discovery for oxide perovskite catalysts

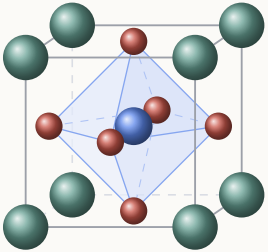


Oxide perovskite ABO_3: a BO_6 octahedron in a cage of A cations

Oxide perovskites catalyse the **oxygen-evolution reaction** (OER), the bottleneck of water electrolysis for renewable hydrogen.

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A-site cation *B*-site O

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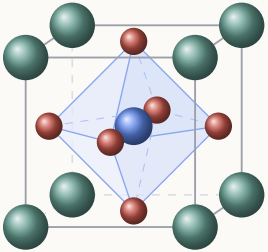
Oxide perovskites catalyse the **oxygen-evolution reaction** (OER), the bottleneck of water electrolysis for renewable hydrogen.

Which mix of **size**, **charge** and **electronegativity** makes a perovskite a good catalyst?

$$t = \frac{R_A + R_O}{\sqrt{2}\,(R_B + R_O)} \qquad \mu = \frac{R_B}{R_O} \qquad Q_A \quad N_d \quad \chi_A \quad \chi_B$$

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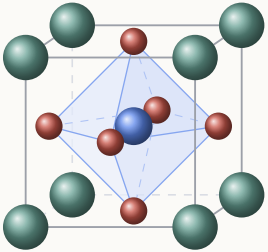
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Goal: a **materials gene**, a short, interpretable *descriptor* equation linking these factors to activity, *with* a measure of how sure we are.

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A probabilistic framework motivated by this descriptor-discovery problem. (We return to it in Part 6.)

2

Symbolic Regression

2 The SR problem: learn the equation itself

Given noisy observations

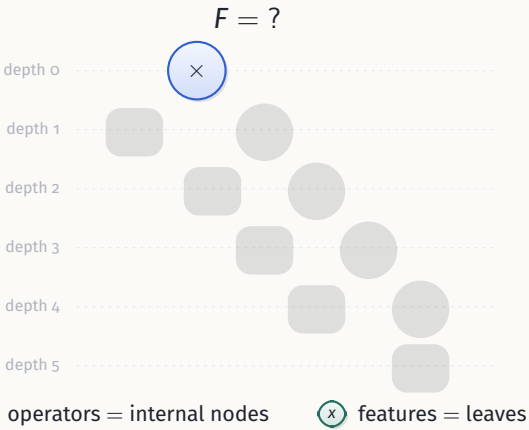
$$y_i = f_0(\mathbf{x}_i) + \epsilon_i, \qquad i = 1, \ldots, n,$$

learn the unknown relationship f_0.

- **Regression** : a predictor of f_0.
- **SR** : its algebraic structure, composed from

$$\{x_1, \ldots, x_p;\ +, \times, \sin, \exp, \ldots\} \Longrightarrow \widehat{f}(\mathbf{x}).$$

Example: the Lorentz force on a moving charge, built one node at a time.



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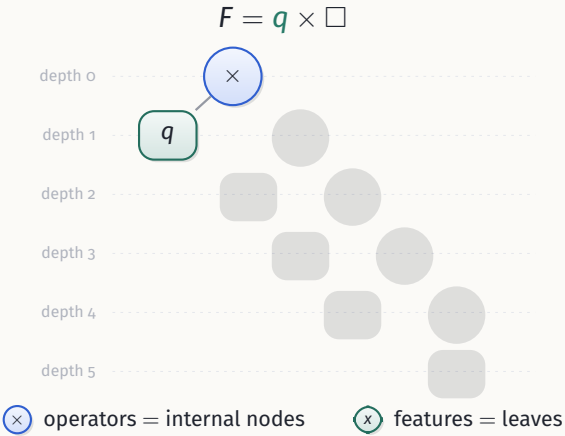
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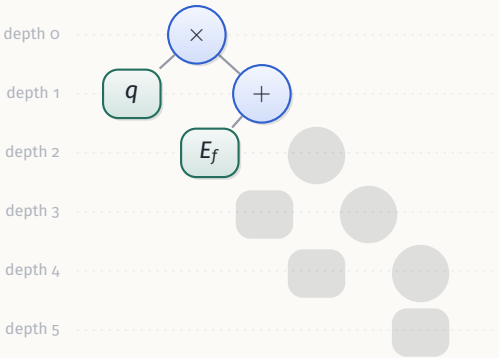
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$$F = q\,(E_f + \square)$$



⊗ operators = internal nodes ⓧ features = leaves

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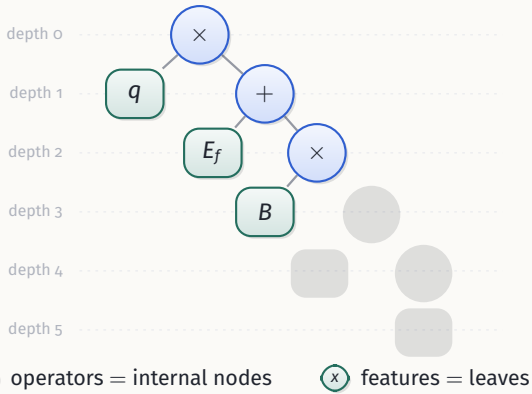
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$$F = q\,(E_f + B \times \square)$$



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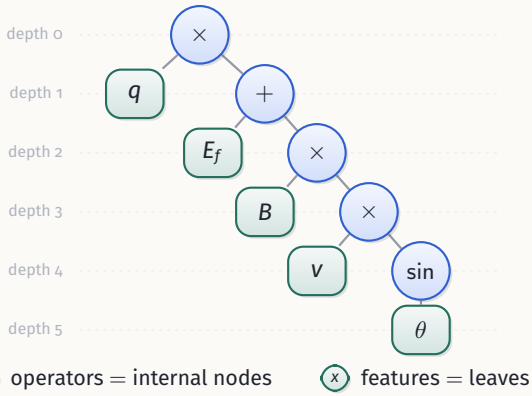
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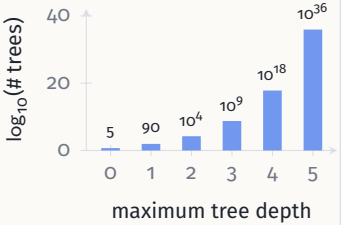
$$F = q\,(E_f + B\,v\,\sin\theta)$$



Every equation has a **symbolic tree** representation:
operators at the internal nodes, features at the leaves.

2 Challenges in symbolic regression

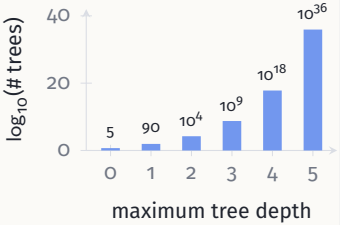
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9 operators, 5 features.
[Virgolin & Pissis, *TMLR* (2022)]:
NP-hard

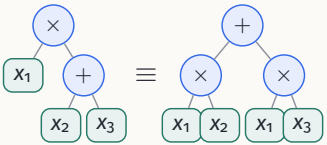
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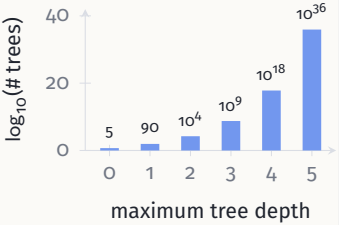


$$x_1(x_2 + x_3) = x_1x_2 + x_1x_3$$

Different trees, identical predictions:
the *formula* is not identifiable from data.

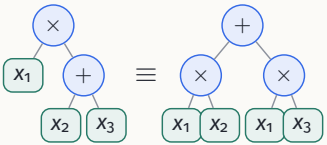
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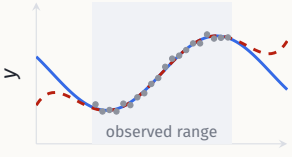
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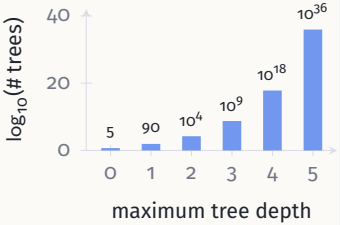


$\sin x$ $\quad x - \frac{x^3}{6} + \frac{x^5}{120}$

Both fit the data; they disagree outside it.

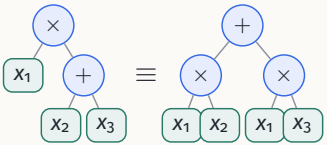
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We need **accuracy**, **parsimony**, **recovery** *and* an honest account of **symbolic uncertainty**.

2 Symbolic regression landscape

A shared benchmark: SRBench [La Cava et al., *NeurIPS* (2021); Imai Aldeia et al., *GECCO* (2025)]



Evolutionary
mutate & select

gplearn
Stephens (2016)

Operon
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PySR
Cranmer (2023)

genetic programming,
since Koza (1992)

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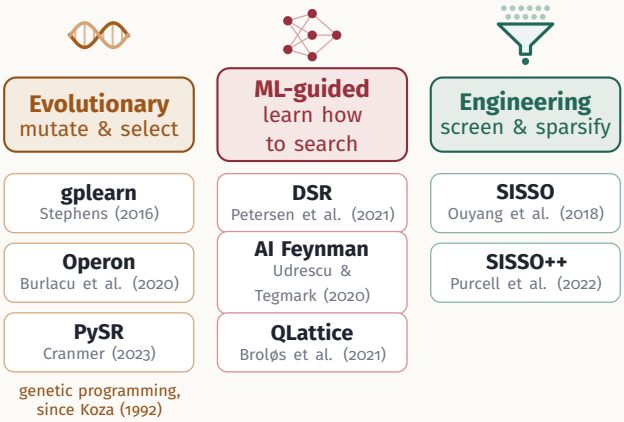
ML-guided
learn how to search

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LLM-guided
LLMs propose

- **LaSR** Grayeli et al. (2024)
- **LLM-SR** Shojaee et al. (2025)

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- **BSR** Jin et al. (2020)
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focus of this talk



LLM-guided
LLMs propose

- **LaSR** Grayeli et al. (2024)
- **LLM-SR** Shojaee et al. (2025)

2 Symbolic regression landscape

A shared benchmark: SRBench [La Cava et al., *NeurIPS* (2021); Imai Aldeia et al., *GECCO* (2025)]

Each family makes a different trade-off

Evolutionary	Flexible and mature; compute cost and expression bloat need control.
ML-guided	Fast, amortized search; favours large samples, low noise, regular structure.
Engineering	Exhaustive and sparse; relies on huge candidate libraries.
Bayesian	Uncertainty-aware in principle; plug-in coefficients, fixed operator weights, single-equation outputs.
LLM-guided	Brings domain knowledge; sensitive to prompts and the language model [Roy et al., *UAI* (2026)].

2 Probabilistic symbolic regression

The lens of this talk: a posterior over equations, not one best fit

$$\underbrace{\Pi(f \mid \mathcal{D}_n)}_{\text{posterior over equations}} \propto \underbrace{p(\mathcal{D}_n \mid f)}_{\text{how well } f \text{ explains the data}} \times \underbrace{\Pi(f)}_{\text{which formulas are plausible a priori}}$$

	BSR Jin et al. (2020)	**BMS** Guimerà et al. (2020)	HierBOSSS this talk
Symbolic representation	tree ensemble	single tree	
Coefficient learning	least-squares plug-in	n/a	
Operator / feature preferences	fixed (user)	fixed (corpus)	
Tree-specific weights	✗	n/a	
Active component selection	✗	n/a	
Symbolic uncertainty summary	✗	✗	

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Operator / feature preferences	fixed (user)	fixed (corpus)	**data-adaptive**
Tree-specific weights	✗	n/a	✓
Active component selection	✗	n/a	✓
Symbolic uncertainty summary	✗	✗	✓

2 Probabilistic symbolic regression

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Tree-specific weights	✗	n/a	✓
Active component selection	✗	n/a	✓
Symbolic uncertainty summary	✗	✗	✓

HierBOSSS: a **fully probabilistic** SR framework that learns structure, controls complexity, and reports *several* plausible laws.

3

Introducing HierBOSSS

3 HierBOSSS at a glance

HierBOSSS: Hierarchical Bayesian Operator-induced Symbolic regression trees for Structural learning of Scientific expressions

1

Symbolic forest



K trees, affine combination

HierBOSSS: Hierarchical Bayesian Operator-induced Symbolic regression trees for Structural learning of Scientific expressions

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Symbolic forest



K trees, affine combination

2

Regularized prior



shallow trees; penalize complex o/p

HierBOSSS: Hierarchical Bayesian Operator-induced Symbolic regression trees for Structural learning of Scientific expressions

1 Symbolic forest



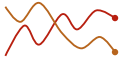
K trees, affine combination

2 Regularized prior



shallow trees; penalize complex o/p

3 Posterior sampling



collapsed MCMC over forests, parallel chains

HierBOSSS: Hierarchical Bayesian Operator-induced Symbolic regression trees for Structural learning of Scientific expressions

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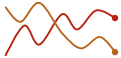
K trees, affine combination

2 Regularized prior



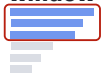
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collapsed MCMC over forests, parallel chains

4 Occam's window



keep the top-*r* forests by posterior score

HierBOSSS: Hierarchical Bayesian Operator-induced Symbolic regression trees for Structural learning of Scientific expressions

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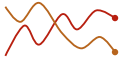
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5 Refinement



BIC symbolic refinement $\rightarrow$ *Final* laws

HierBOSSS: Hierarchical Bayesian Operator-induced Symbolic regression trees for Structural learning of Scientific expressions

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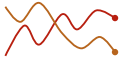
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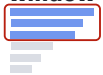
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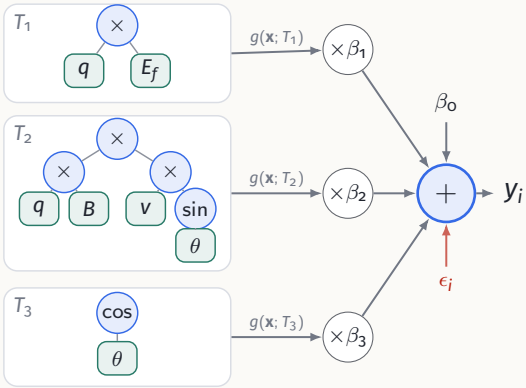


BIC symbolic refinement → *Final* laws

Output: **a set of plausible equations**, each with posterior support, rather than one best guess.

The symbolic forest model

An additive regression model whose components are formulas



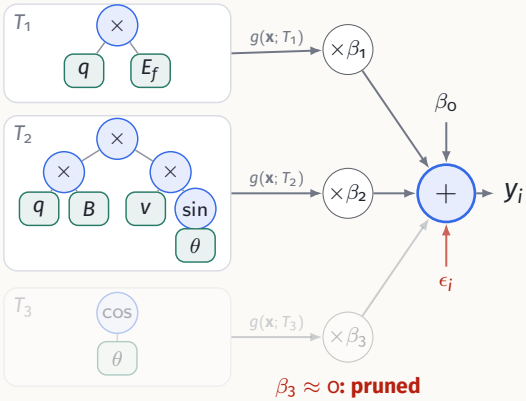
$$y_i = \beta_0 + \sum_{j=1}^{K} \beta_j \, g(\mathbf{x}_i; T_j) + \epsilon_i$$

$$\epsilon_i \sim \mathrm{N}(0, \sigma^2)$$

- $g(\mathbf{x}; T_j)$: evaluate tree T_j at $\mathbf{x}$; weight it by β_j; **add** the components.
- Each tree is one interpretable **component**. Example: Lorentz force with $K = 3$.

The symbolic forest model

An additive regression model whose components are formulas



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- Each tree is one interpretable **component**. Example: Lorentz force with $K = 3$.
- Weak or collinear trees are removed later: $K_{\text{eff}} = 2 \leq K$.

3 A regularizing prior: Occam's razor

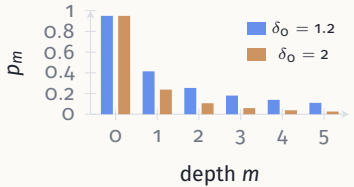
Favour simple formulas [Jefferys & Berger, *American Scientist* (1992)]

SPLIT PROBABILITY AT DEPTH m

$$p_m = \alpha_0 (1+m)^{-\delta_0}$$

$$\alpha_0 \in (0,1), \delta_0 > 0$$

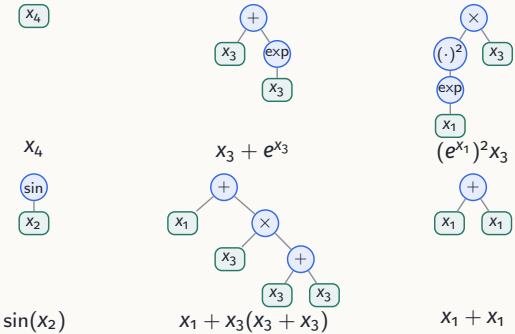
[Chipman et al., *JASA* (1998)]



$\alpha_0 = 0.95$. Deep nodes rarely split: long formulas are *possible* but must earn their keep.

SIX CONSECUTIVE DRAWS FROM THE HierBOSSS TREE PRIOR

$\delta_0 = 1.2$; operators $+, \times, \sin, \exp, \mathrm{inv}, (\cdot)^2$; features $x_1, \ldots, x_4$



3 A regularizing prior: Occam's razor

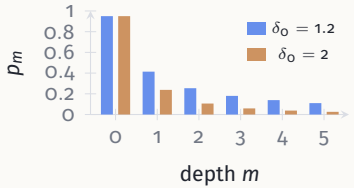
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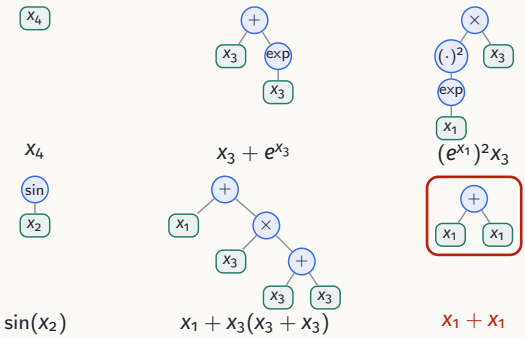
[Chipman et al., *JASA* (1998)]



$x_1 + x_1$ says the same as x_1 (up to β). **Among equivalent encodings, the shorter one gets more prior mass.**

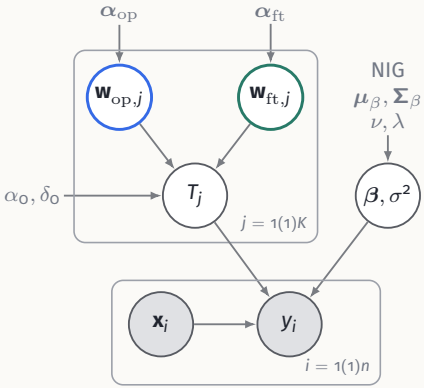
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3 Data-adaptive Dirichlet weights

The full hierarchical model

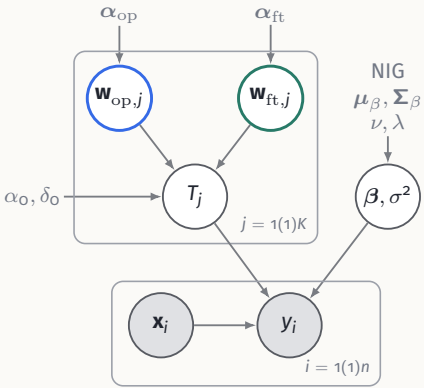


Shaded: observed. Plates: repeated for each tree / observation.

Operators $\mathbf{w}_{\text{op},j} \sim \text{Dir}(\boldsymbol{\alpha}_{\text{op}})$, **features** $\mathbf{w}_{\text{ft},j} \sim \text{Dir}(\boldsymbol{\alpha}_{\text{ft}})$, one pair **per tree**.

Data-adaptive Dirichlet weights

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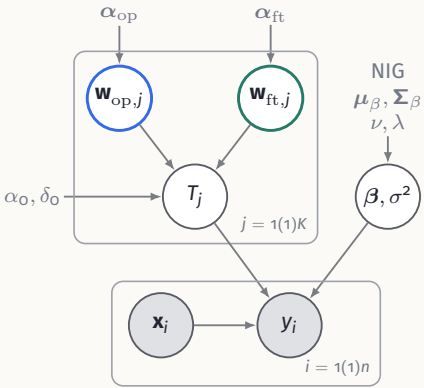
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Integrating the weights out:

$$\Pi_{\text{tree}}(T_j) \propto \underbrace{\mathfrak{B}(\boldsymbol{\alpha}_{\text{op}} + \boldsymbol{\xi}_j)}_{\text{operators}} \underbrace{\mathfrak{B}(\boldsymbol{\alpha}_{\text{ft}} + \boldsymbol{\varrho}_j)}_{\text{features}} \times \underbrace{\prod_m p_m^{|\mathcal{N}_{\text{op},m}|} (1 - p_m)^{|\mathcal{N}_{\text{ft},m}|}}_{\text{depth penalty}}$$

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Preferences for operators and features are **learned from the data**, tree by tree.

Key distinction from BMS (fixed, corpus-based) and BSR (fixed, user-specified).

Occam's razor, quantified

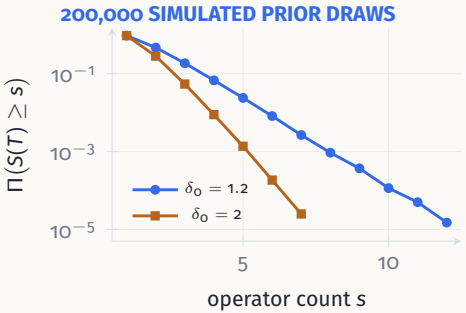
Proposition (tail control of the prior)

For every depth d and operator count s,

$$\Pi_{\text{tree}}\big(\mathrm{d}(T) \geq d\big) \leq C_1 e^{-c_1 d \log d},$$
$$\Pi_{\text{tree}}\big(S(T) > s\big) \leq C_2 e^{-c_2 s},$$

and the whole forest obeys the same exponential bound.

- Complex formulas get **exponentially small** prior mass.
- This is exactly what the **theory** needs later: little prior mass outside a sieve of bounded complexity.



Straight lines on a log scale = exponential decay.

The collapsed target

Integrate out (β, σ^2) by Normal–Inverse-Gamma conjugacy

$$\mathrm{JMP}(\mathcal{T}) \propto \underbrace{\det\left(\boldsymbol{\Sigma}_\beta^\star(\mathcal{T})\right)^{1/2}\left\{\lambda^\star(\mathcal{T})\right\}^{-\nu^\star/2}}_{\text{marginal likelihood of the forest}} \times \underbrace{\Pi_{\mathrm{forest},K}(\mathcal{T})}_{\text{Occam prior}}$$

$$\boldsymbol{\Sigma}_\beta^\star(\mathcal{T}) = \left(\boldsymbol{\Sigma}_\beta^{-1} + \mathcal{E}^\top\mathcal{E}\right)^{-1}, \quad \boldsymbol{\mu}_\beta^\star(\mathcal{T}) = \boldsymbol{\Sigma}_\beta^\star\left(\boldsymbol{\Sigma}_\beta^{-1}\boldsymbol{\mu}_\beta + \mathcal{E}^\top\mathbf{y}\right), \quad \nu^\star = \nu + n,$$
$$\lambda^\star(\mathcal{T}) = \lambda + \mathbf{y}^\top\mathbf{y} + \boldsymbol{\mu}_\beta^\top\boldsymbol{\Sigma}_\beta^{-1}\boldsymbol{\mu}_\beta - \boldsymbol{\mu}_\beta^{\star\top}\boldsymbol{\Sigma}_\beta^{\star-1}\boldsymbol{\mu}_\beta^\star, \qquad \mathcal{E} = \mathcal{E}(\mathbf{X};\mathcal{T}) = \left[\mathbf{1},\ g(\mathbf{X};T_1),\ldots,g(\mathbf{X};T_K)\right]$$

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Score any forest
in closed form: fit and complexity in one number

Rank forests
JMP drives the Occam's window summary

Sample faster
MCMC moves only on the discrete trees

3 The full posterior sampler

Metropolis-within-partially collapsed Gibbs [van Dyk & Park, *JASA* (2008)]

for $\text{iter} = 1, \ldots,$ `max_iter`:

1. **Trees** (target JMP) **for** $j = 1, \ldots, K$:
   propose $T_j^\star$ by one of 7 local moves, π_{move}
   accept with probability

$$\min\left\{1,\ \frac{\text{JMP}(\mathcal{T}^\star)}{\text{JMP}(\mathcal{T})}\,\frac{q(T_j^\star \to T_j)}{q(T_j \to T_j^\star)}\right\}$$

2. **Coefficients and noise** (exact draw)
   $(\boldsymbol{\beta}, \sigma^2) \mid \mathcal{T}, \mathcal{D}_n \sim \text{NIG}\left(\boldsymbol{\mu}_\beta^\star, \boldsymbol{\Sigma}_\beta^\star, \nu^\star, \lambda^\star\right)$

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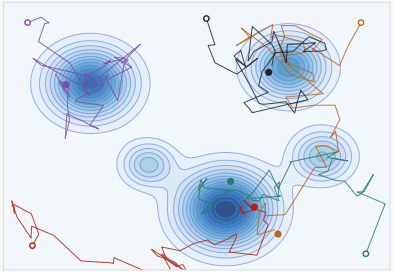
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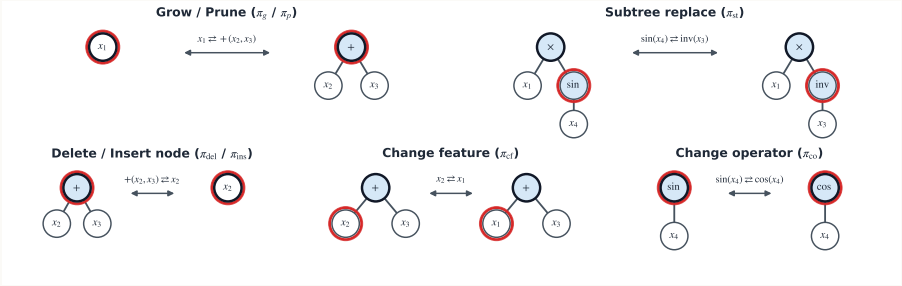


Run **several chains in parallel** from random forests: the space of formulas is **multimodal**.

Schematic: chains on a toy multimodal target.

Seven local tree moves

Step 1 of the sampler: how a candidate tree $T_j^\star$ is proposed



grow / prune · replace a subtree · delete / insert a node · change a feature · change an operator

reversible edits ⇒ a valid Metropolis–Hastings chain over the discrete space of formulas.

Occam's window and refinement

From thousands of samples to a few laws

Pool chains
all visited forests

→

Rank by JMP
highest first

→

Occam's window
$\mathcal{J}_r = \{\mathcal{T}^{(1)}, \ldots, \mathcal{T}^{(r)}\}$

[Madigan & Raftery, *JASA* (1994)]

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→

Refine each forest
BIC subset, refit β, simplify (SymPy)

$K_{\text{eff}} \leq K$ trees kept

→ ***Final***

REAL OUTPUT: LORENTZ FORCE, NOISE $\sigma = 1$ (REPETITION 1)

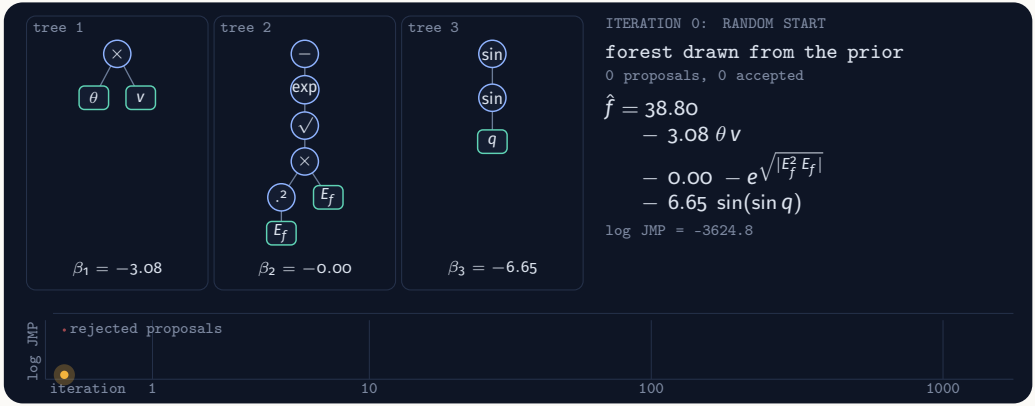
1 2 3 4 5 6 7 8 9 10 → $\widehat{F} = 1.003\, qE_f + 0.999\, qBv \sin\theta$

all $r = 10$ forests in the window

Every forest refines to the *same* law: here the posterior is confident. Truth: $F = q(E_f + Bv \sin\theta)$.

3 HierBOSSS in action: from a prior draw to the law

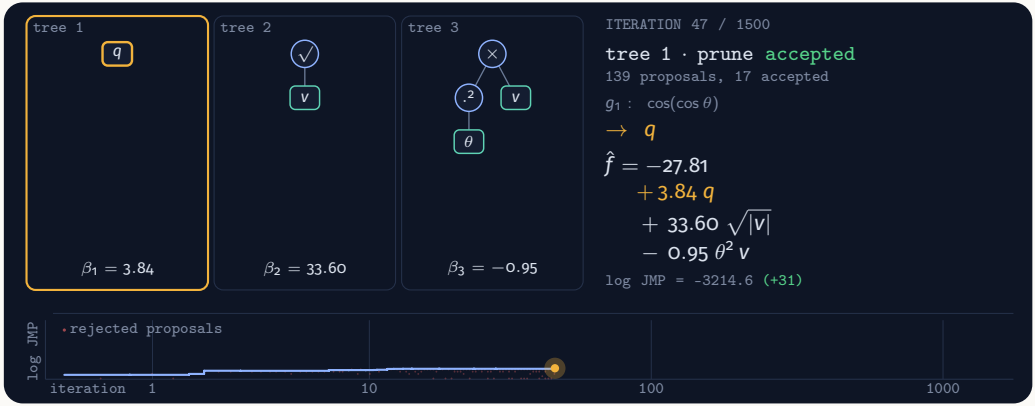
One real chain (seed 101 of 10) on the Lorentz force law, $F = q\,(E_f + Bv\sin\theta)$



The **regularized prior** proposes forests, **collapsed MCMC** keeps the moves the data support, and **Occam's window + BIC** returns the law with its alternatives.

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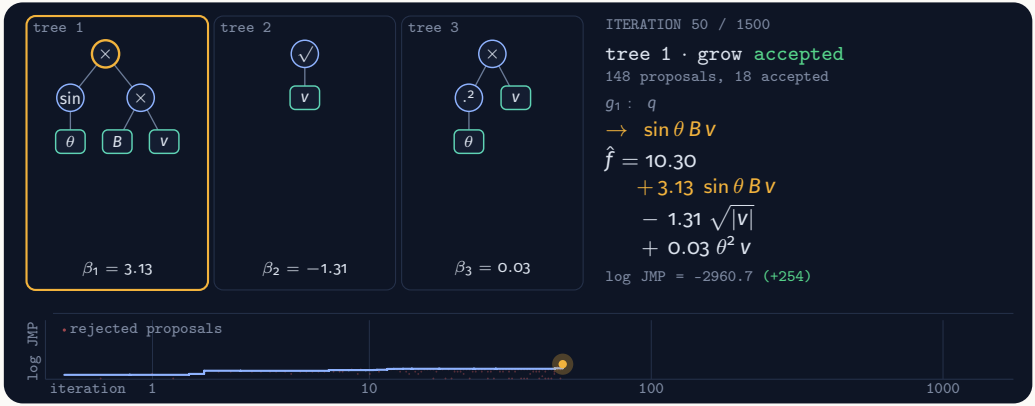
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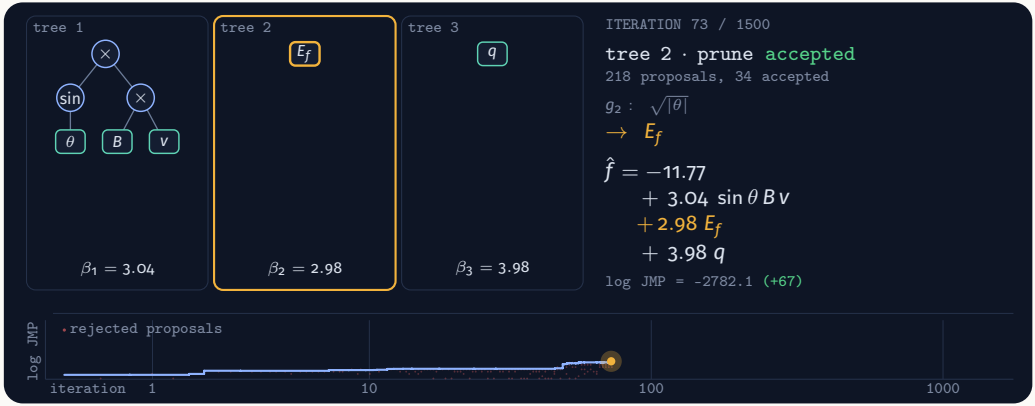
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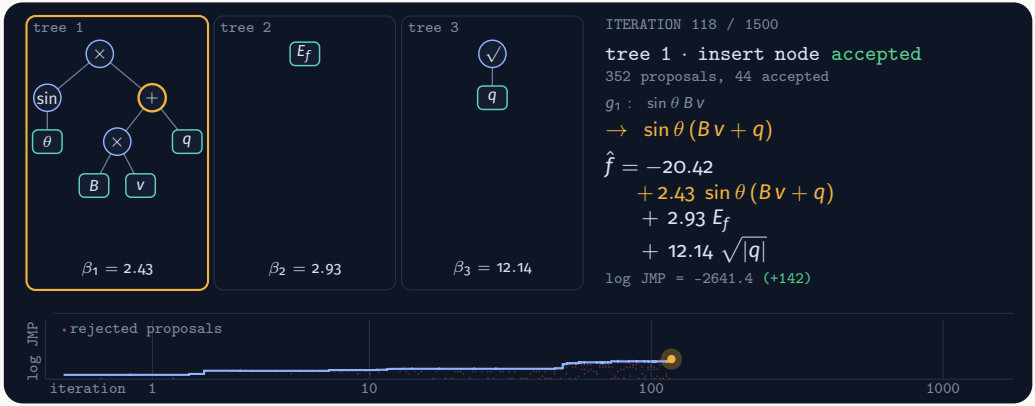
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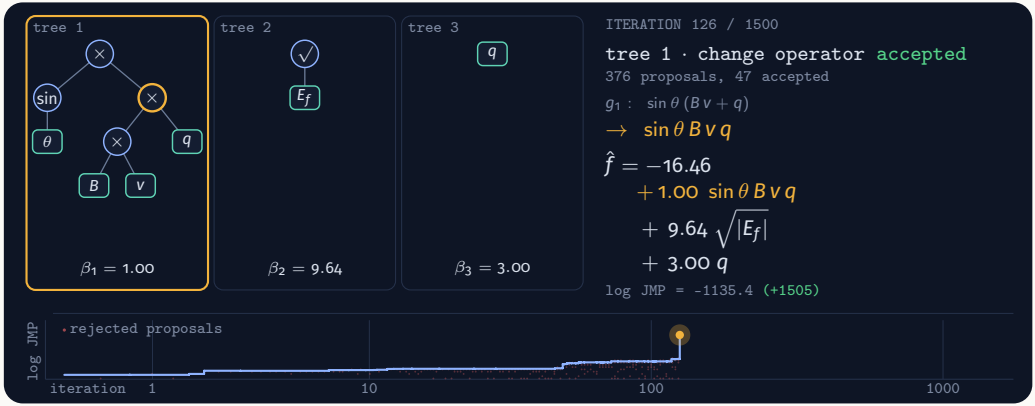
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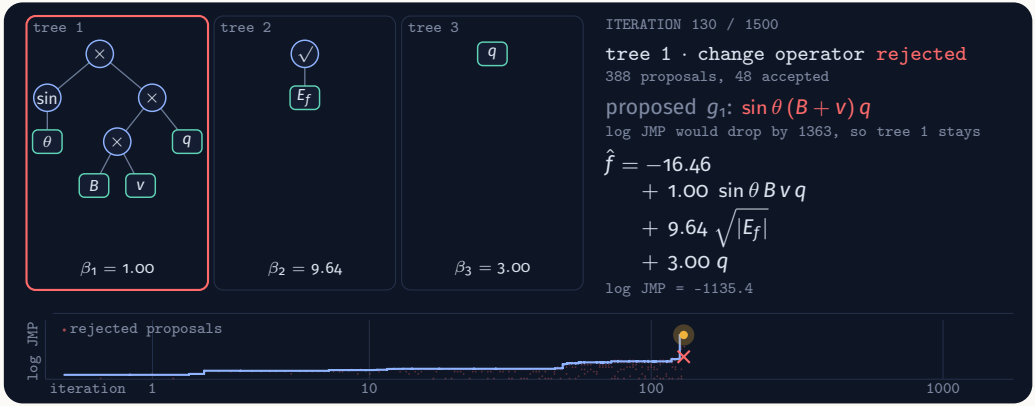
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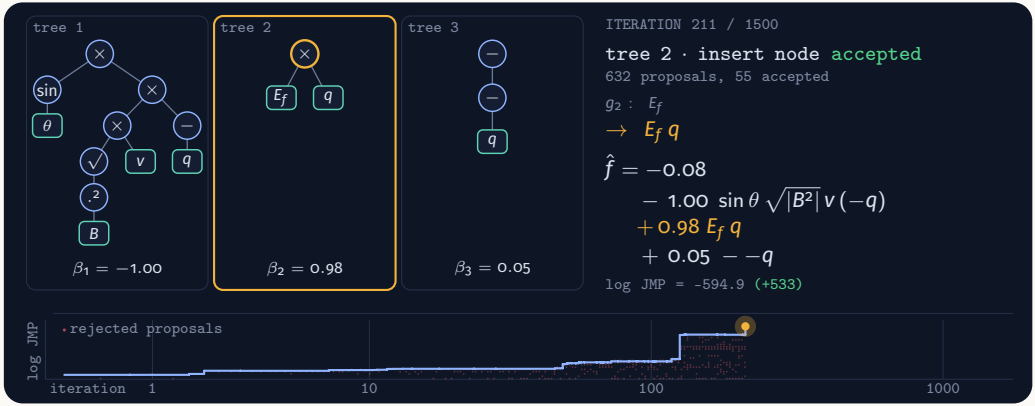
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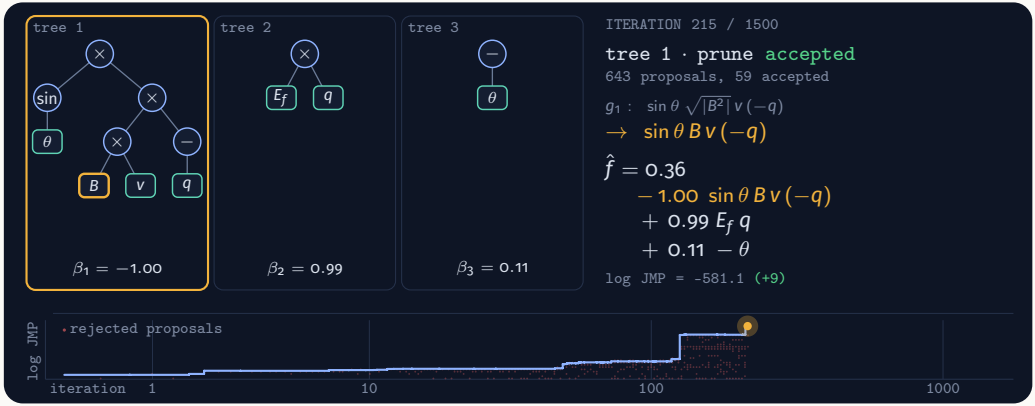
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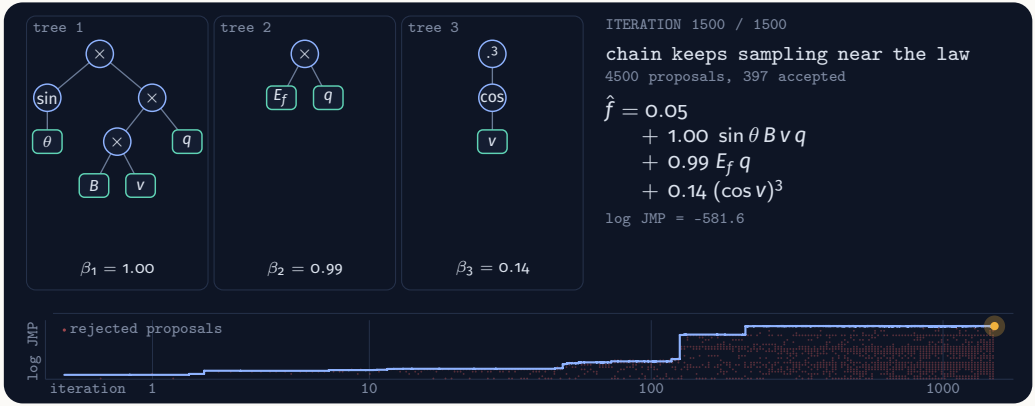
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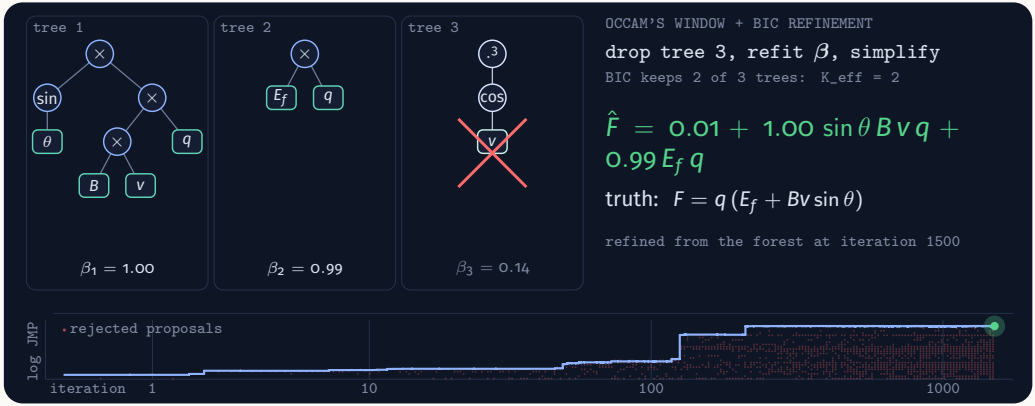
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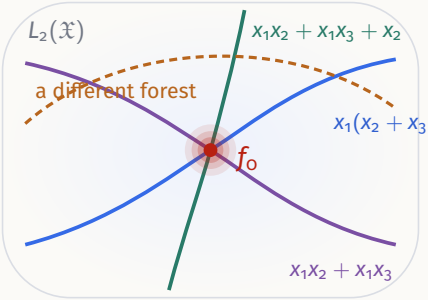


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4

What The Posterior Learns

4 What can a theory of symbolic regression promise?



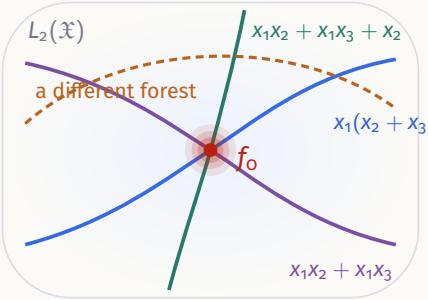
Each curve: the functions one forest structure produces as its coefficients vary. Curves through f_0: *different formulas, same function.*

THE SYMBOLIC FUNCTION CLASS

$$\mathcal{F}_K = \left\{ f_{\boldsymbol{\beta},\mathcal{T}}(\mathbf{x}) = \beta_0 + \sum_{j=1}^{K} \beta_j \, \mathrm{ev}[T_j](\mathbf{x}) \right\}$$

with $\mathcal{F}_{K,S} \subset \mathcal{F}_K$: forests with at most S operators.

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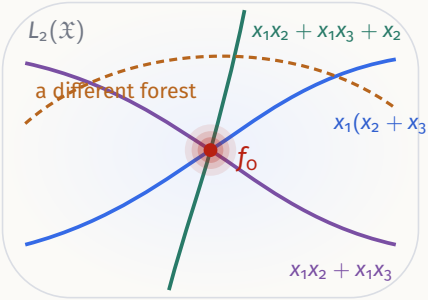
with $\mathcal{F}_{K,S} \subset \mathcal{F}_K$: forests with at most S operators.

- The likelihood sees f only at the design points, so distances are

$$d_n^2(f, g) = \tfrac{1}{n} \textstyle\sum_{i=1}^{n} \{f(\mathbf{x}_i) - g(\mathbf{x}_i)\}^2.$$

- Guarantees concern the **function**, not a unique formula: equivalent formulas keep posterior mass.

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This is *why* `HierBOSSS` reports an **Occam's window** of formulas rather than a single winner.

4 The price of symbolic expressivity

One number, $\mathfrak{C}_{K,S,n}$, puts discrete search and continuous estimation on one scale

$$\mathfrak{C}_{K,S,n} = \underbrace{S\,[1 + \log(S + |\mathbb{O}|)]}_{\text{tree shapes}} + \underbrace{(S + K)\,[1 + \log(p + S + K)]}_{\text{node labels}}$$
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How many forest structures have S operators in total.

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How many forest structures have S operators in total.

Node labels
Which operator or feature sits at each node.

Size of evaluations
Operators such as exp and $\times$ can blow values up: assumption (SE).

Coefficients & noise
Resolution needed for β and σ^2.

$\mathfrak{C}_{K,S,n}$ acts as an **effective dimension** of symbolic search: it controls metric entropy and prior mass.

4 Theorem 1: when the law is reachable

Approximate symbolic realizability (ASR): richer forests approximate f_0 arbitrarily well

Theorem 1 (posterior concentration)

Fix $K, p, \mathbb{O}$. Under (SE), (PR) and (ASR), for some $M > 0$,

$$\mathrm{E}_{\mathbb{P}_0^n} \Pi\big(d_n(f, f_0) > M\, r_{n,K} \,\big|\, \mathcal{D}_n\big) \longrightarrow 0,$$

where, up to a logarithmic factor,

$$n\, r_{n,K}^2 \asymp \inf_{S \in \mathbb{N}} \Big\{ \underbrace{n\, a_{K,S,n}^2(f_0)}_{\text{approximation}} + \underbrace{\mathfrak{C}_{K,S,n}}_{\text{complexity}} \Big\}.$$

$a_{K,S,n}(f_0) = \inf_{f \in \mathcal{F}_{K,S}} d_n(f, f_0)$: best error with operator budget S.

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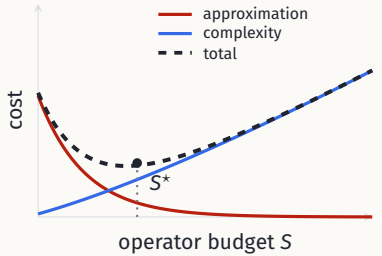
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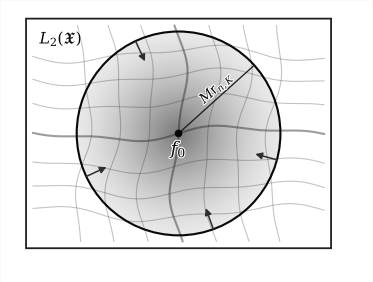
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Schematic. The posterior finds its own best budget.

The rate is the **best trade-off** between approximating f_0 and paying for expressivity.

4 Exact laws are learned at a near-parametric rate



Posterior mass collapses onto a shrinking ball around f_0.

Corollary (exact symbolic realizability)

If f_0 *is* a finite formula, $f_0 \in \mathcal{F}_{K,S_0}$, then

$$n\, r_{n,K}^2 \le \mathfrak{C}_{K,S_0,n}\left[1 + \log(\mathfrak{C}_{K,S_0,n} + K + p + |\mathbb{O}|)\right],$$

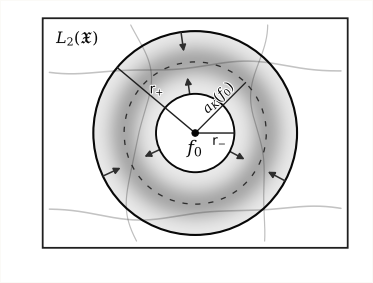
and $\mathfrak{C}_{K,S_0,n} = \mathcal{O}(\log n)$.

$$r_{n,K} = \mathcal{O}\left(n^{-1/2}\sqrt{\log n \, \log\log n}\right)$$

Almost as fast as a **parametric** model, even though the space of formulas is infinite.

4 Theorem 2: when no formula is exactly right

Symbolic misspecification: f_0 may lie outside $\mathcal{F}_K$



Mass concentrates on a thin *annulus* whose radius is the best achievable error.

Best achievable (population) error, possibly *never attained*:

$$a_K^2(f_0) = \inf_{f \in \mathcal{F}_K} \int |f - f_0|^2 \, q_X.$$

Theorem 2 (sharp oracle concentration)

Under (SE), (PR) and bounded coefficients (BC), the sieve-truncated posterior puts mass $\to 1$ on

$$\left\{ f : \left| d_n^2(f, f_0) - a_K^2(f_0) \right| \lesssim \mathfrak{q}_{n,K} \right\}, \qquad \mathfrak{q}_{n,K} \to 0.$$

$$d_n^2(f, f_0) \;\le\; \mathbf{1} \cdot a_K^2(f_0) + \mathcal{O}(\mathfrak{q}_{n,K})$$

leading constant exactly one

4 Theorem 2: when no formula is exactly right

Symbolic misspecification: f_o may lie outside $\mathcal{F}_K$

Best achievable (population) error, possibly *never*

Why this is new

- **No pseudo-true formula needed.** $\mathcal{F}_K$ may have *no* best formula, or infinitely many equivalent ones.
- **No testing over convex covers.** Classical misspecification theory [Kleijn & van der Vaart, *AoS* (2006)] needs these; they are hard to verify for a discrete, nonconvex class of formulas.
- A **sharp oracle** statement in the sense of [Dalalyan & Salmon, *AoS* (2012)]: the price of the wrong model class is paid exactly once.

Mass concentrates on a thin *annulus* whose radius is the best achievable error.

$$d_n^2(f, f_o) \leq \mathbf{1} \cdot a_K^2(f_o) + \mathcal{O}(q_{n,K})$$

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5

Rediscovering Physics

5 Can we rediscover physics from noisy data?

Five laws from the Feynman Symbolic Regression Database [Udrescu & Tegmark, *Sci. Adv.* (2020)]

Coulomb's law CL

$$F = \frac{q_1 q_2}{4\pi\epsilon r^2}$$

$p = 4$ · low complexity

Lorentz force FCE

$$F = q(E_f + Bv\sin\theta)$$

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Forced oscillator EHFO

$$E_n = \tfrac{1}{4}m(\omega^2 + \omega_0^2)x^2$$

$p = 4$ · medium complexity

Nonlinear oscillator HOQN

$$x = x_1[\cos\omega t + \alpha\cos^2\omega t]$$

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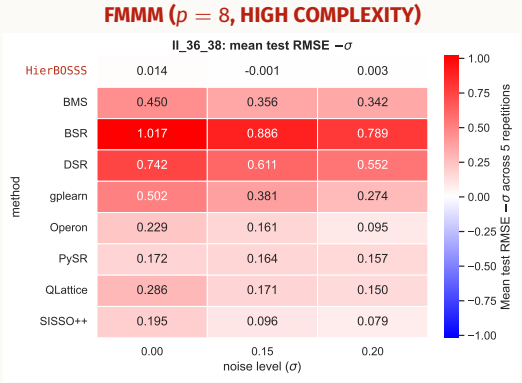
8 SRBench competitors | $n = 2000$ **per run** | **5 repetitions, 90/10 split** | **noiseless + 2 noise levels**

gplearn, Operon, PySR | DSR, QLattice | SISSO++ | BMS, BSR. Shared operator library

$\mathbb{O} = \{+, \times, \mathrm{neg}, \mathrm{inv}, \sin, \cos, \exp, (\cdot)^2, (\cdot)^3\}$ (SISSO++: a per-law subset).

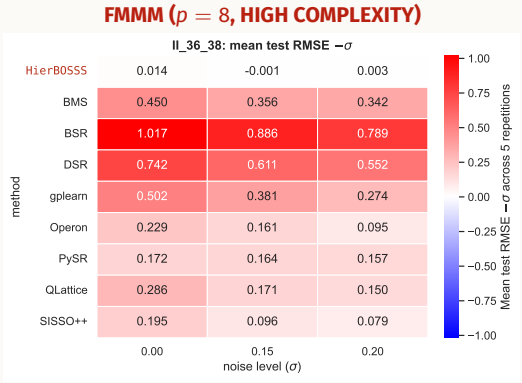
5 Accuracy: the hardest law separates the methods

Mean test RMSE minus noise level; ≈ 0 means predicting at the noise floor



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HierBOSSS

≤ 0.014

eight competitors

$0.079 - 1.017$

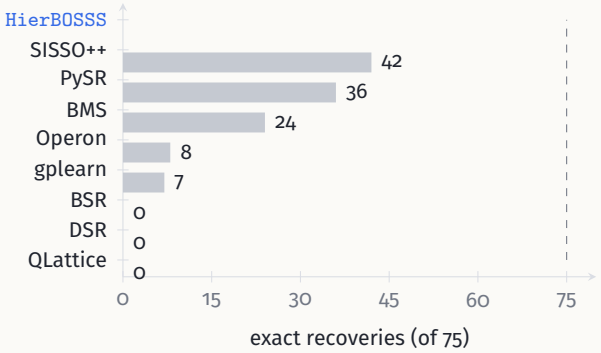
best competitor: SISSO++ (0.079 at $\sigma = 0.20$)

On the easier laws (FCE, EHFO) the field is closer: several methods sit near the noise floor (backup).

Good prediction is necessary. But does it find the *right law*?

5 Prediction is not recovery

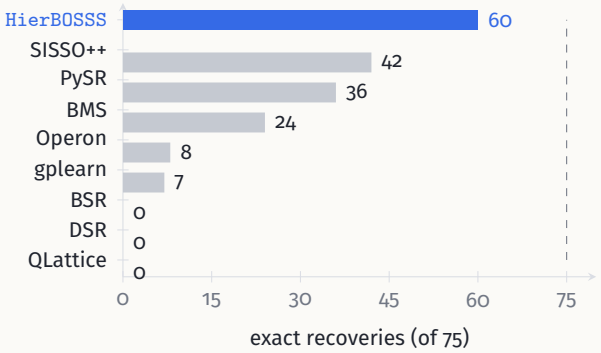
Exact symbolic recoveries out of 75 runs (5 laws × 3 noise settings × 5 repetitions)



Totals of the paper's per-law recovery heatmap (backup).

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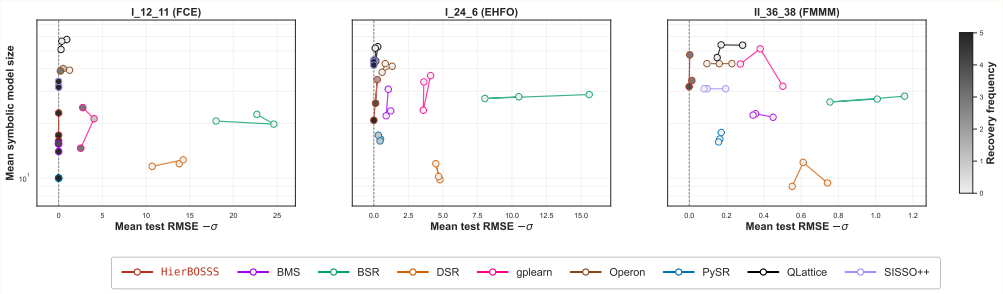


Totals of the paper's per-law recovery heatmap (backup).

- **Accurate ≠ correct:** BMS, Operon, PySR and QLattice often predict well but recover inconsistently.
- Only `HierBOSSS` recovers the hardest law: **FMMM in 10 of 15 runs**.
- PySR leads on HOQN; SISSO++ is strong when the law lies inside its enumerated space (FCE, EHFO).

5 Accuracy, complexity and recovery, together

Lorentz force (FCE), forced oscillator (EHFO) and magnetic moment (FMMM)



← lower test error ↓ smaller expression darker = more recoveries

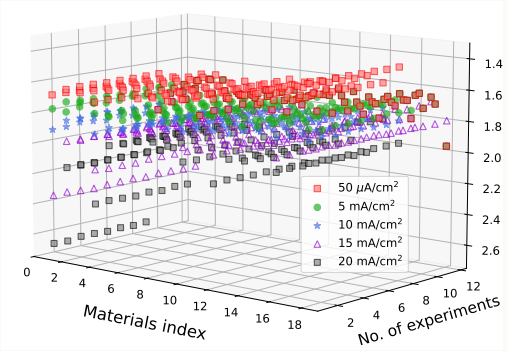
HierBOSSS sits in the **low-error, low-complexity, high-recovery** corner, and runs considerably faster than BMS and BSR (backup).

6

Discovery of Materials Genes

6 Back to the perovskites: data and target

Experimental data of [Weng et al., *Nature Communications* (2020)]



Measured potential V_{RHE} across 18 catalysts, replicates and five current densities ρ.
Lower V_{RHE} = more active.

18 ABO_3 catalysts $n = 1080$ $p = 8$

4 preparations × 3 repeats × 5 current densities

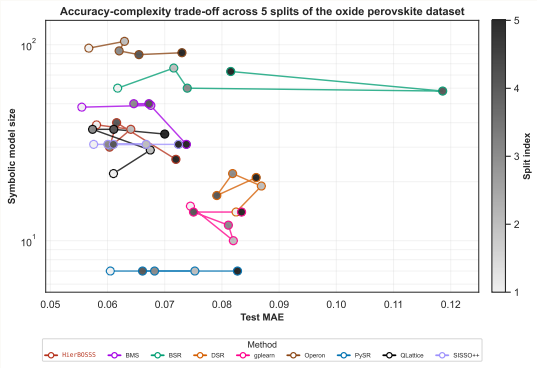
SYMBOLIC DISCOVERY TARGET

$$V_{\mathrm{RHE}} = f(\rho, \mathbf{z}) + \epsilon,$$
$$\mathbf{z} = (R_A,\ Q_A,\ N_d,\ \chi_A,\ \chi_B,\ \mu,\ t)$$

Operators $\{+, \times, \mathrm{neg}, \mathrm{inv}, \sqrt{\cdot}\}$ for every method; nominal $K = 8$ from an ablation (backup).

6 Accurate *and* compact

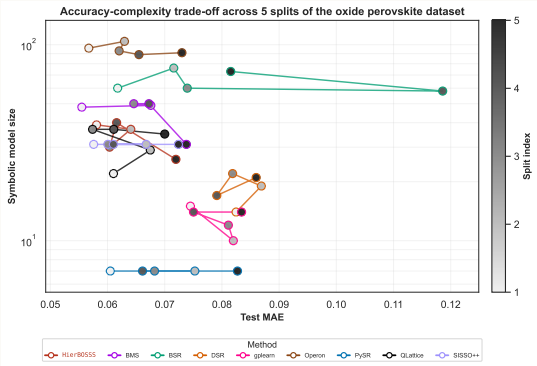
Held-out error vs. expression size over 5 train/test splits



Read toward the **lower left**: small error, small formula.

6 Accurate *and* compact

Held-out error vs. expression size over 5 train/test splits



Read toward the **lower left**: small error, small formula.

HierBOSSS
test MAE 0.058–0.072 eV
26–40 nodes

Accurate, not compact: Operon 89–104 nodes, BSR 58–76.

Compact, less accurate: PySR 7 nodes, gplearn 10–15.

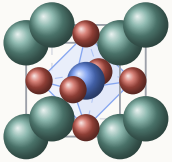
HierBOSSS repeatedly sits near the **accuracy–complexity Pareto frontier** (BMS and QLattice are competitive on some splits).

6 The recovered materials gene

TOP-RANKED DESCRIPTOR (SPLIT 5)

$$\widehat{V}_{\mathrm{RHE}} = 0.371\, t + \frac{1.288}{t} + \frac{0.334\,\rho}{N_d} + 0.014\, N_d\, \rho - 0.069\, \chi_B\, \rho$$

test MAE 0.072 **eV** $K_{\mathrm{eff}} = 5$ **of** $K = 8$ **26 nodes**



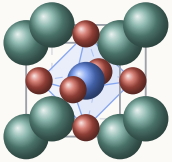
t: geometric fit of A, B, O
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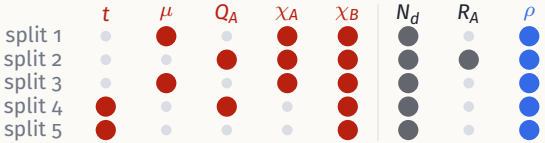
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DESCRIPTORS USED BY THE TOP FORMULA, PER SPLIT



key OER descriptors in Weng et al. (2020)

HierBOSSS recovers the descriptor families known to drive OER activity, plus a dependence on the current density ρ.

6 Symbolic uncertainty is real

Occam's window ($r = 10$ top forests), coloured by the distinct *Final* formula each one refines to



	distinct laws	test MAE (eV)
Lorentz force	**1**	(same law)

When the data pin down one law, the window collapses (Feynman). When they do not, `HierBOSSS` **says so**.

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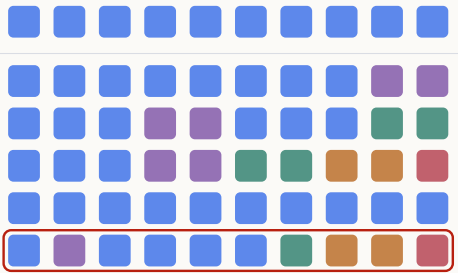


	distinct laws	test MAE (eV)
Lorentz force	1	(same law)
Perovskite split 1	2	0.057–0.058
Perovskite split 2	3	0.063–0.064
Perovskite split 3	5	0.059–0.063
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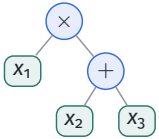
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Split 5: all five laws share the core $1/t + \rho/N_d + N_d\rho + \chi_B\rho$ and differ in *one* extra term (t, $\sqrt{t}$, $1/\chi_A$, $\sqrt{R_A}$ or a constant).

7

Outlook

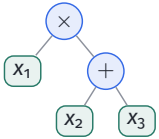
7 Takeaways



Discover

Learn **compact symbolic structure**: forests of operator-induced trees under a regularized prior.

7 Takeaways



Discover

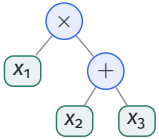
Learn **compact symbolic structure**: forests of operator-induced trees under a regularized prior.



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Report a **set of plausible laws** (Occam's window), not one fitted equation.

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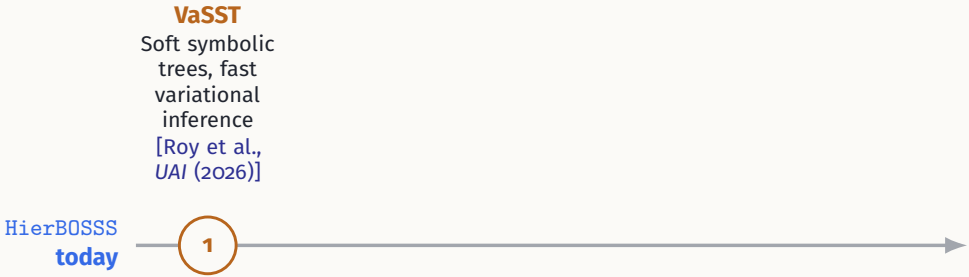
Trust

Posterior concentration: near-parametric for exact laws, sharp oracle under misspecification.

One probabilistic framework: discover the law, quantify the alternatives, and know when to trust it.

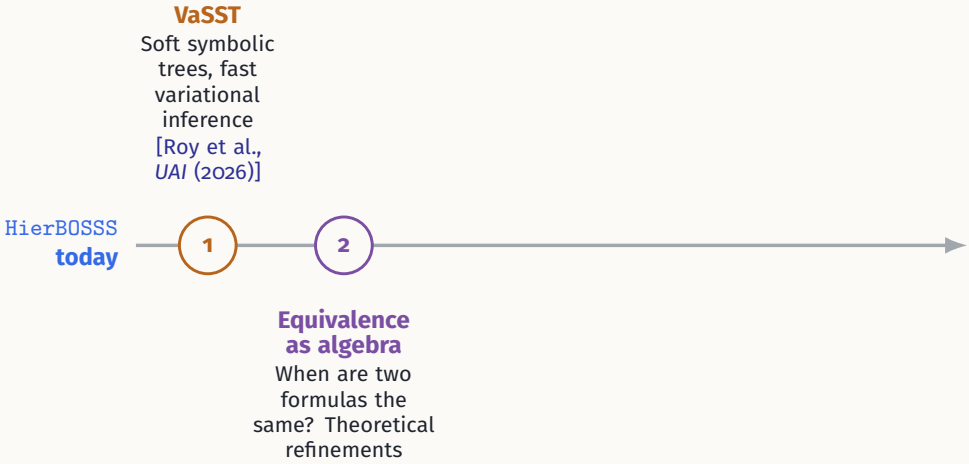
7 Toward self-driving probabilistic discovery

Future directions: what we are working on



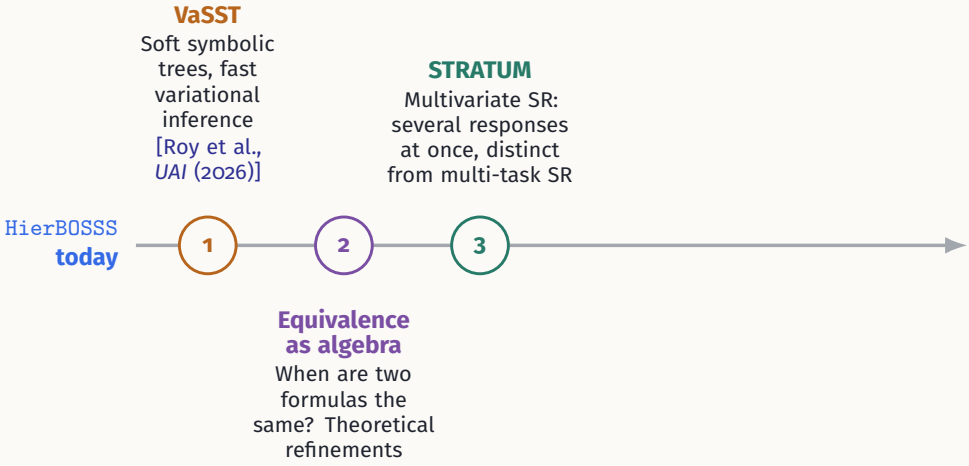
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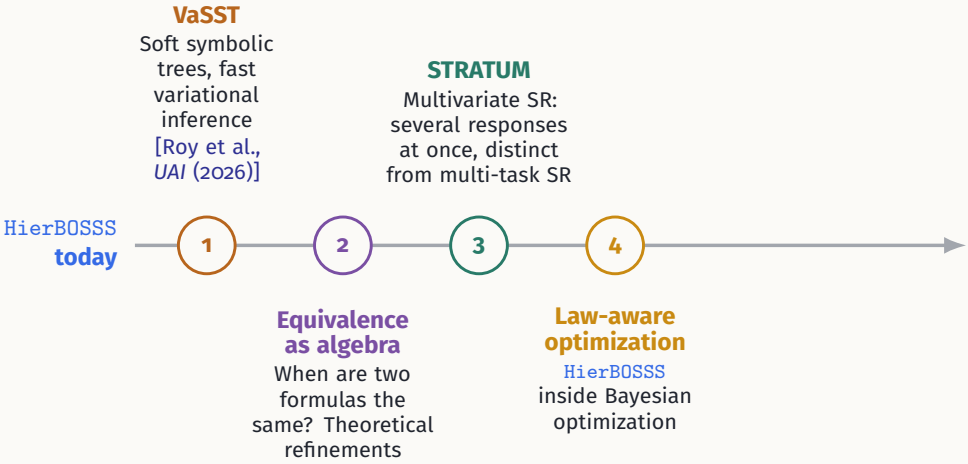
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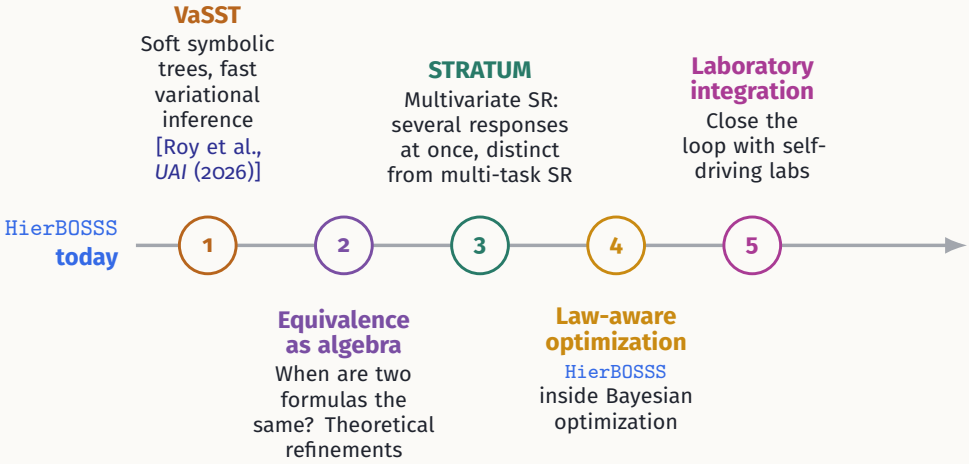
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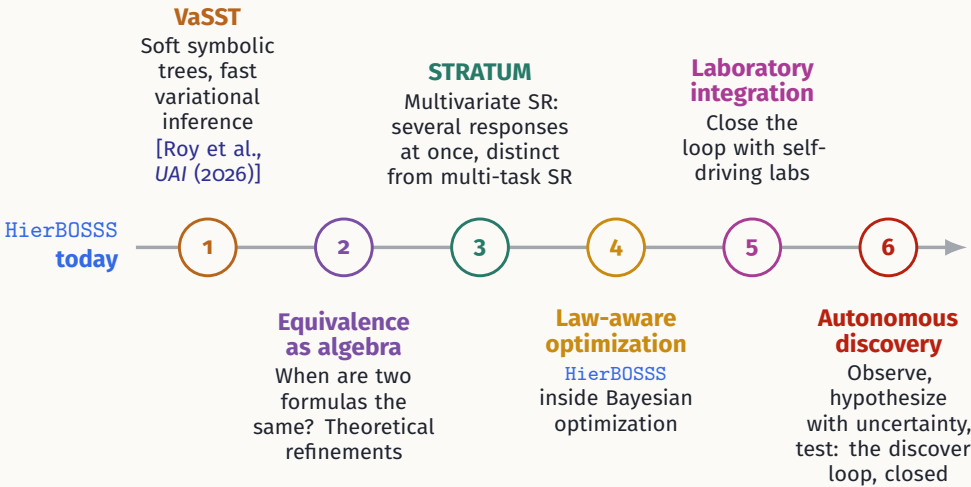
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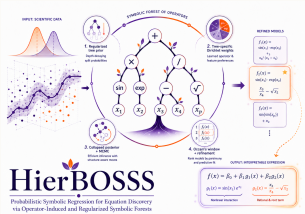


Thank you!

Obrigado!

Questions and discussion are welcome.

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Texas A&M University
sroy_123@tamu.edu



Paper on arXiv

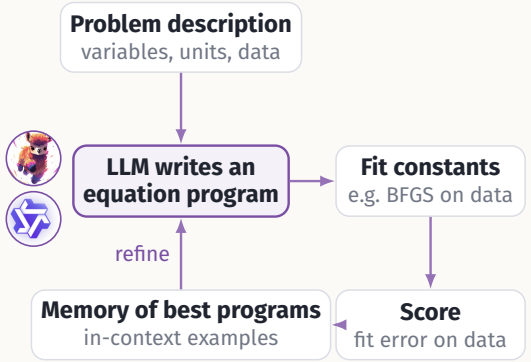
Code on GitHub

A

Backup

A LLM-guided SR: language models as equation proposers

Examples of the newest family in the landscape



AN EQUATION PROGRAM (illustrative, LLM-SR style)

```
def equation(x, v, params):
    """Acceleration of a damped,
    nonlinear oscillator."""
    return (- params[0] * x
            - params[1] * v
            - params[2] * x**3)
```

The LLM chooses the *structure*; the optimizer chooses the *numbers*.

LLM-SR equations as Python programs + an experience buffer [Shojaee et al., *ICLR* (2025)]

LaSR LLM-written "concept library" steers PySR's evolutionary search [Grayeli et al., *NeurIPS* (2024)]

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What LLMs add

- ✔ **Scientific priors** learned from text
- ✔ **Programs**, not just trees
- ✔ **Feedback loops** that refine hypotheses

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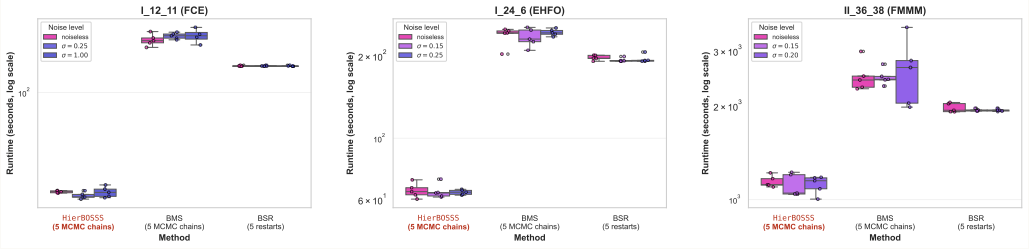
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`HierBOSSS` is complementary: **explicit, data-adaptive priors** and a **posterior over equations**.
A natural next step: LLM-informed priors inside a Bayesian SR posterior.

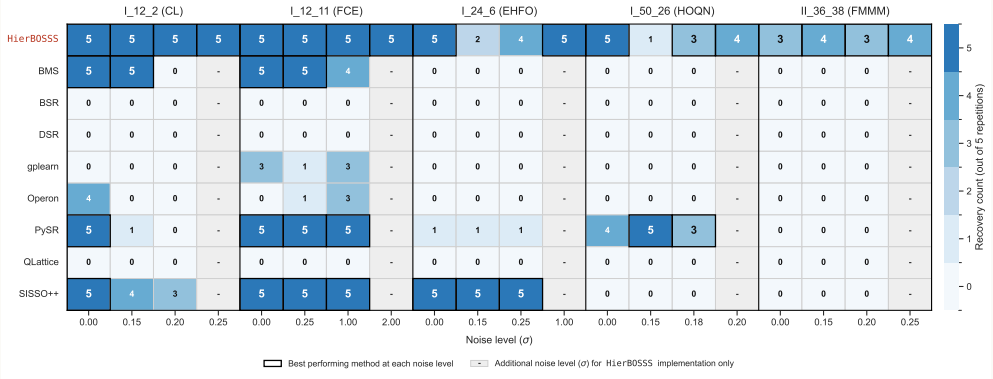
A Runtime against the Bayesian SR modules



Wall-clock time (log scale). Five MCMC chains for HierBOSSS and BMS; five restarts for BSR.

HierBOSSS is considerably faster than BMS and BSR across low-, medium- and high-complexity laws.

A Exact recovery, per law and noise level

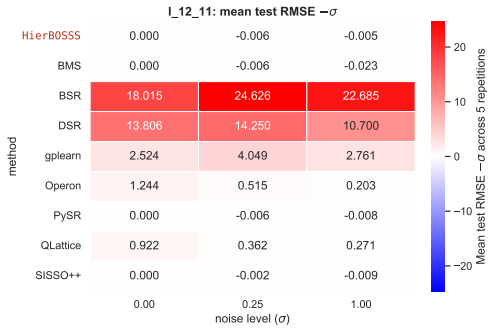


Count out of 5 repetitions. The rightmost column of each law is an extra, HierBOSSS-only stress test (not included in the totals of the main slide).

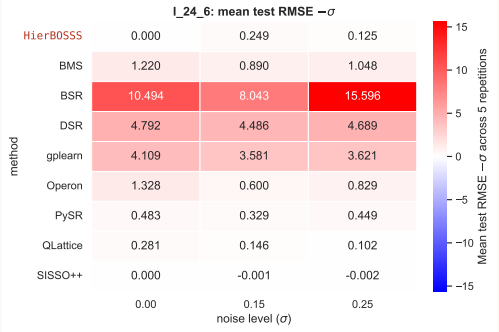
A Accuracy on the easier laws

Mean test RMSE minus noise level, averaged over 5 repetitions

FCE (LORENTZ FORCE)

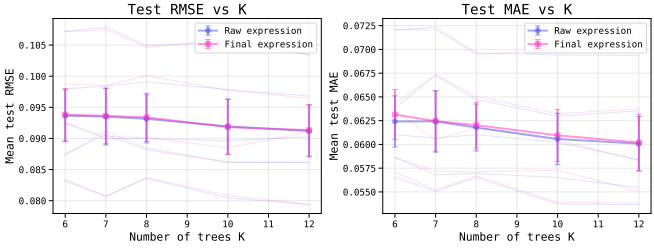


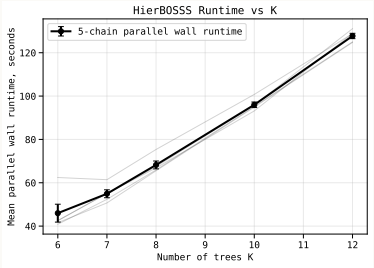
EHFO (FORCED OSCILLATOR)



A How many trees? Choosing the nominal K

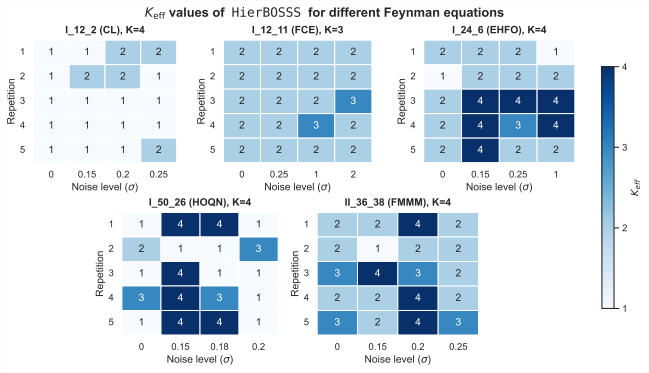
Perovskites: held-out error and runtime across forest sizes





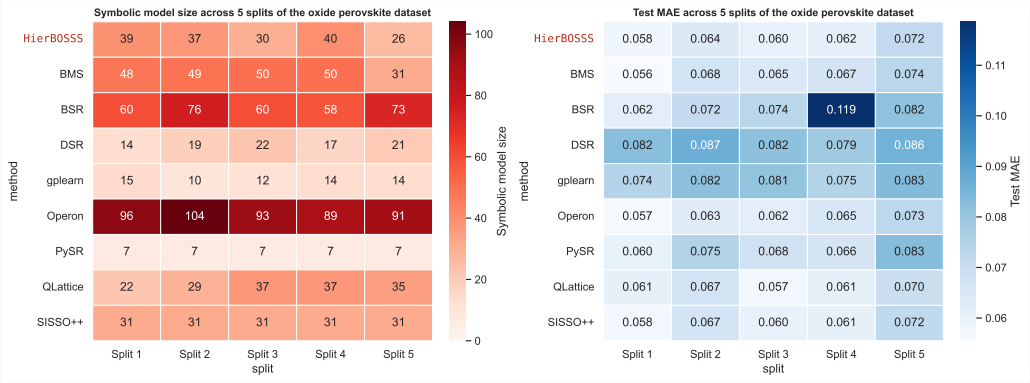
Held-out error stabilizes around K=8 trees; larger forests only add runtime.

A Effective forest size K_{eff} on the Feynman laws

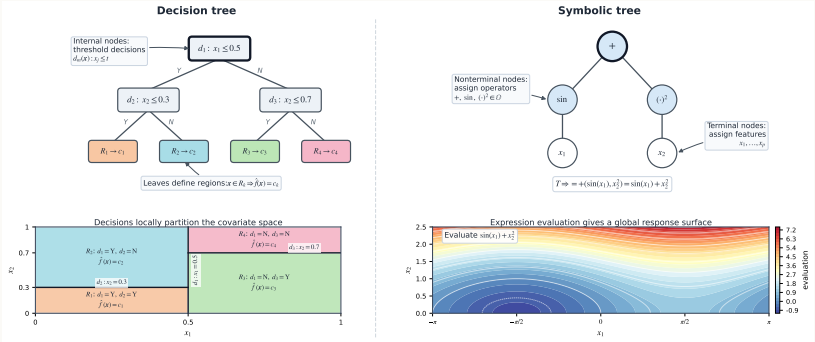


BIC-based refinement often keeps fewer trees than the nominal K for simple laws, and 3–4 components for structurally richer ones.

A Model size and error by split (perovskites)



A Decision trees vs. symbolic trees



Decision trees split the input space into regions with constant leaves [Chipman et al., *JASA* (1998); Denison et al., *Biometrika* (1998)]. Symbolic trees compose operators and features into one global formula.

A Assumptions behind the theorems

(SE) **Evaluation envelope.** $\sup_{\mathbf{x}\in\mathfrak{X},\, S(T)\le S} |\mathrm{ev}[T](\mathbf{x})| \le U_S \le \bar{C}_U\, e^{c_U S}$.

(PR) **Prior regularity.** Dirichlet parameters bounded: $0 < a_{\mathrm{op}} < \alpha_{\mathrm{op},o} < A_{\mathrm{op}}$, $0 < a_{\mathrm{ft}} < \alpha_{\mathrm{ft},h} < A_{\mathrm{ft}} < \infty$.

(ASR) **Approximate symbolic realizability.** For every S there is $f_{(S)} \in \mathcal{F}_{K,S}$ with $d_n(f_{(S)}, f_0) \to 0$ as $S \to \infty$.

(BC) **Bounded coefficients.** $|f_0| \le R_0$, and some sequence $f_m \in \mathcal{F}_K$ with $\|\beta_m\|_\infty \le C_\beta$ attains $\mathsf{E}_{q_X}|f_m - f_0|^2 \to a_K^2(f_0)$.

Theorem 2 uses a design-dependent sieve $\mathcal{F}_n^D = \{S(\mathcal{T}) \le S_n,\ \|\beta\|_\infty \le C_B,\ d_n(f,0) \le C_D\}$ with $S_n \to \infty$; $q_{n,K} = \max\{\rho_{n,K}, \xi_n\}$, where $n\rho_{n,K}^2 = B_{n,K}^4\, \mathfrak{C}_{K,S_n,n}[1 + \log(\mathfrak{C}_{K,S_n,n} + K + p + |\mathbb{O}|)]$. Proofs in the supplement of the paper.